

view displays. But this approach is not successful as, in case of many users, it leads to flickering.

2. Providing multiple views. With multiple views, a proper 3D pattern can be formed.

To create multiple views, the most common method is to use lenticular lenses. Wikipedia defines a lenticular lens as an array of magnifying lenses so that, when viewed from slightly different angles, different images are magnified.

In multi-view lenticular autostereoscopic displays, the LCD is located at the focal plane of lenses (lenticular). As the image in LCD is placed at a focal plane, different view angles become possible. For example, if there are three

columns of pixels, three viewing zones will be possible. But it has a problem of its own. As the number of zones increases, and when the viewer's eyes change the zone, the image appears to flip between the views.

The Stereographics Corporation, USA developed a 106.68cm (42-inch) autostereoscopic monitor that offered nine perspective views.

Holographic approach. Holography is a photographic recording technique that can record and reproduce the image with the properties of light waves, such as amplitude and wavelength, very precisely. In this technique, the scene and the target both are illuminated with coherent light. It is supposed to be a very close

approximate of real 3D viewing. Some basic models have been created that reduce the amount of spatial data.

Pros. True 3D, full parallax.

Cons. Difficulty to receive full colour images, high cost and complicated system.

Holographic optical element displays. These are based on the diffraction property of light. These displays have no information about what is being projected, therefore they just diffract the light. But for this, a tracker is needed that moves the light source in the direction of the movement of head. These displays use a hologram of evenly diffused plane which is in rastered form so that, when the light falls, it would direct it from alternating patterns to specified viewing zones.

Integral imaging. This method was invented by M.G. Lippmann in 1908. Here, a lens array with many convex microlenses is placed in front of a transparent film. This results in a number of small elemental images corresponding to the number of lenses. With these elemental images, the transparent film is developed.

For the reproduction of the object, the transparent film is irradiated from behind by diffused light. This bounds the light beam to retrace the original path and converge at the same point where the real object was. The major advantage of this is that we have several zones of auto stereo vision due to the number of microlenses.

Advantages. Less spatial data required, possibility of simultaneous vision by several viewers and less hardware required.

Problems. The viewing side is opposite to that of pick-up side, thus the resultant image is pseudoscopic in nature with reverse depth, that is, we see a convex image for a concave object. So we would need to convert the pseudoscopic image to orthoscopic.

Besides, there is interference between elemental images during pick-up. This interference can be reduced by setting optical barriers between neighbouring lenses.

The NHK Science & Technology Research Laboratories has developed a hexagonal-packed gradient-index (GRIN) fly-eye lens to capture elemental images. In this process, a convex lens is used to control the depth of the reproduced image. The GRIN structure overcomes the pseudoscopic and interference problem. However, the lens structure that they used was bigger, so the reproduced image was somewhat deteriorated. After lot of research the deterioration has been reduced to an acceptable level.

Overall, we can say that, due to small resolution of displays and charged-coupled devices (CCDs), the depth of 3D is insufficient in this integral imaging.

Volumetric displays. This is a completely different approach as here a 3D

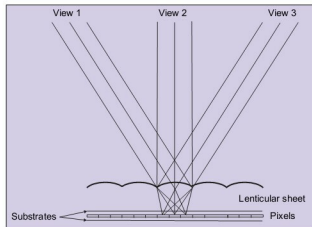


Fig. 6: A multi-view display (Courtesy: www.google.com)

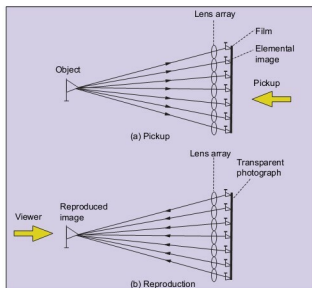


Fig. 7: Integral imaging process (Courtesy: Sergei Dudnikov and Yuri Melnikov)